1. Gas Exchange

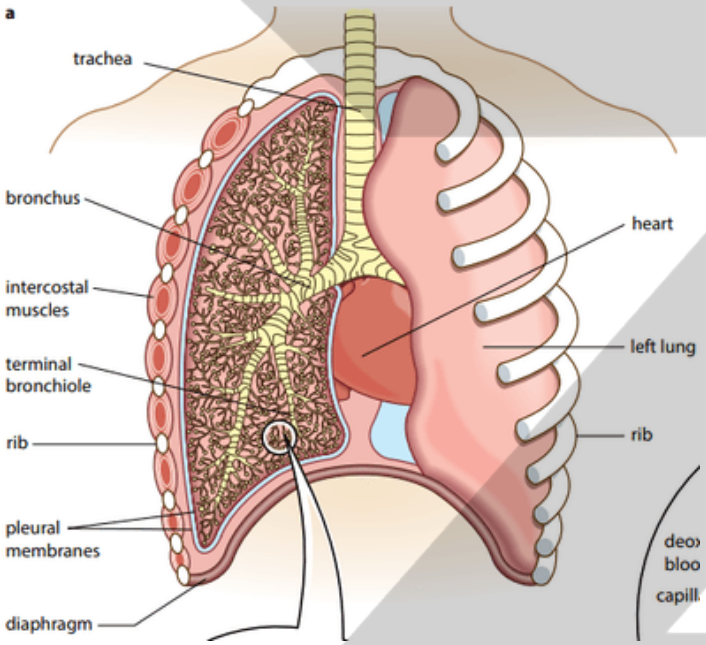
1.1. The Gas Exchange System

Lungs:

- Lungs are in the chest cavity surrounded by the pleural membranes, which enclose an airtight space
- This space contains a small quantity of fluid to allow friction-free movement as the lungs are ventilated by the diaphragm and ribs

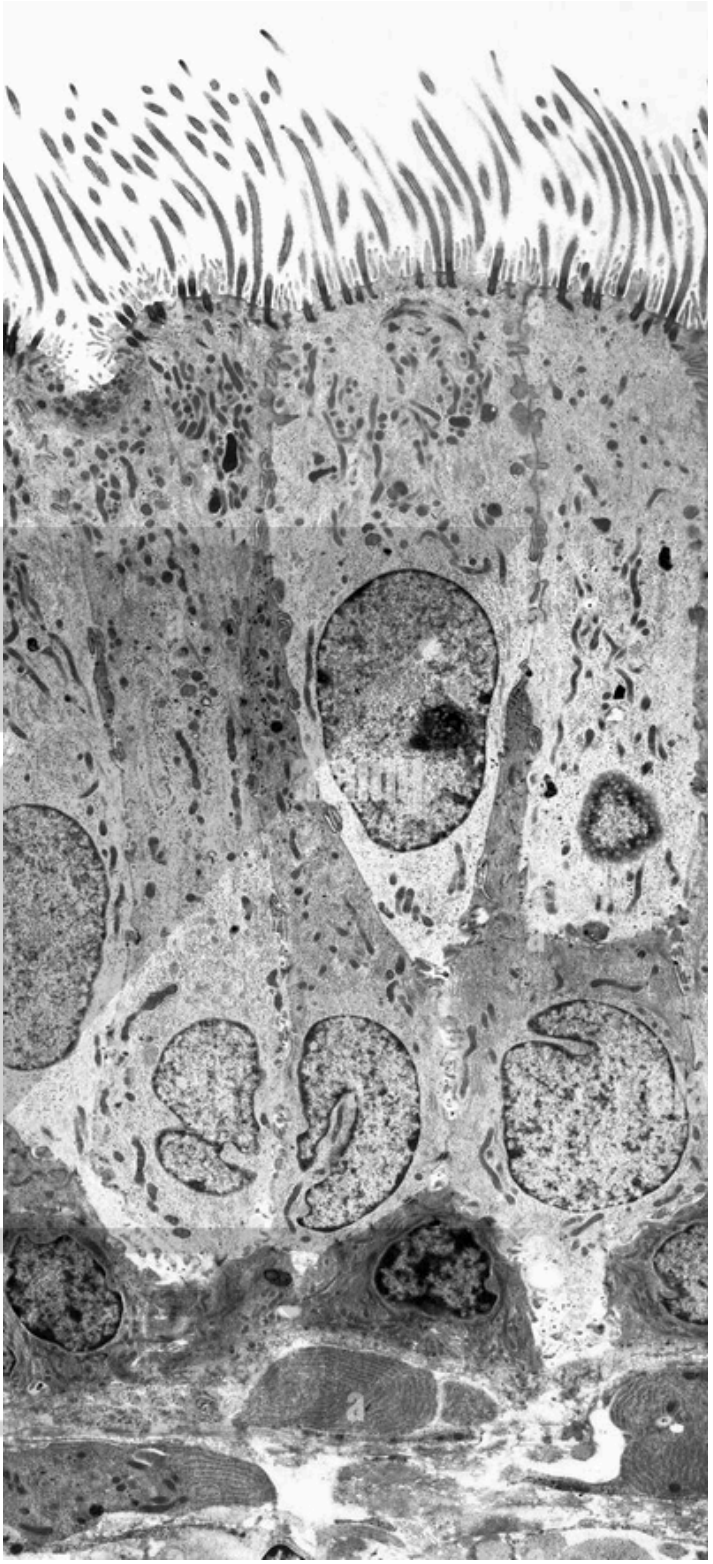
Ventilation of Lungs:

- Replaces alveolar air with air from outside the body
- This helps maintain the diffusion gradient of oxygen and carbon dioxide



Trachea:

- Leading from the throat to the lungs is the trachea
- Cartilage present in regular C-shaped rings
- Ciliated epithelium present, which contains cilia and goblet cells



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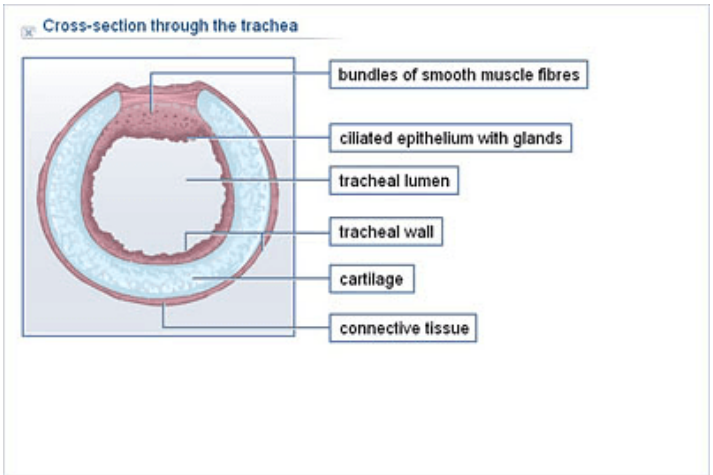
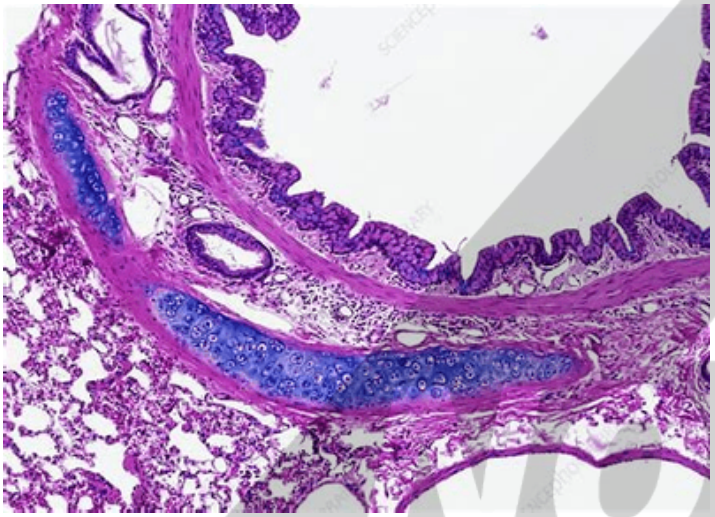
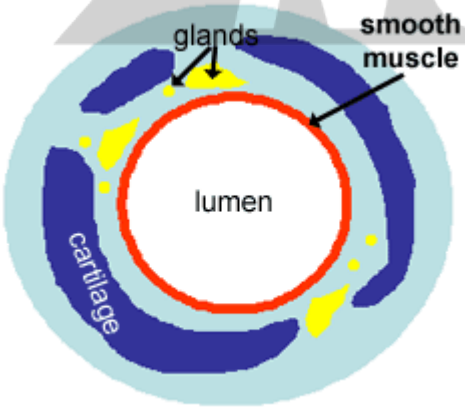


Bronchi:

- Found at the base of the trachea
- Branch to form smaller bronchioles
- Irregular plates of cartilage found
- Ciliated epithelium present
- Surrounded by smooth muscle, which can contract or relax to adjust the diameter of these tiny airways
- During exercise, the muscles relax to allow a greater flow of air to the alveoli.





Cartilage:

- A connective tissue
- It keeps airways in the trachea and bronchi open and air resistance low and prevents them from collapsing or bursting as the air pressure changes during breathing

Epithelium:

- Air flows down the lungs through the trachea and bronchi, which are lined by cells adapted to remove particles from the air before it reaches the lungs
- These cells make up a tissue called epithelium
- There are two main kinds of cells:
  - Ciliated cells: Lined with tiny cytoplasmic extensions known as cilia, they are responsible for the continual beating of mucus toward the larynx
  - Goblet cells: Found in between ciliated cells in large amounts, the upper part of a goblet cell is swollen with mucin droplets that are secreted by the cell; the mucous secreted by goblet cells, traps pathogens which are then moved out with the help of the cilia
    - Mucus: a slimy solution of mucin, which is composed of glycoproteins with many carbohydrate chains that make them sticky and able to trap inhaled particles; it is also made by mucous glands (multicellular) beneath the epithelium in the trachea and bronchus
    - SO_2 and NO_2 can dissolve in mucus to form an acidic solution that irritates the lining of the airways

Bronchioles:

- Smooth muscle present
- Cartilage is absent
- Two kinds of bronchioles- terminal and respiratory
- Respiratory bronchiole leads to the alveoli (site of gas exchange)

Alveoli:

- Alveolar walls contain elastic fibers that stretch during inspiration and recoil during expiration.
- When the alveoli are fully expanded during exercise, the surface area available for diffusion increases
- The walls also contain some collagen fibers to provide support and prevent bursting.
- The walls are extremely thin (only one cell thick)
- Pressed closely against the alveoli walls are blood capillaries (lined by endothelial cells, which are also very thin)
- This allows the diffusion of gases between blood and air in alveoli to take place rapidly
- A large number of alveoli are present to increase surface area

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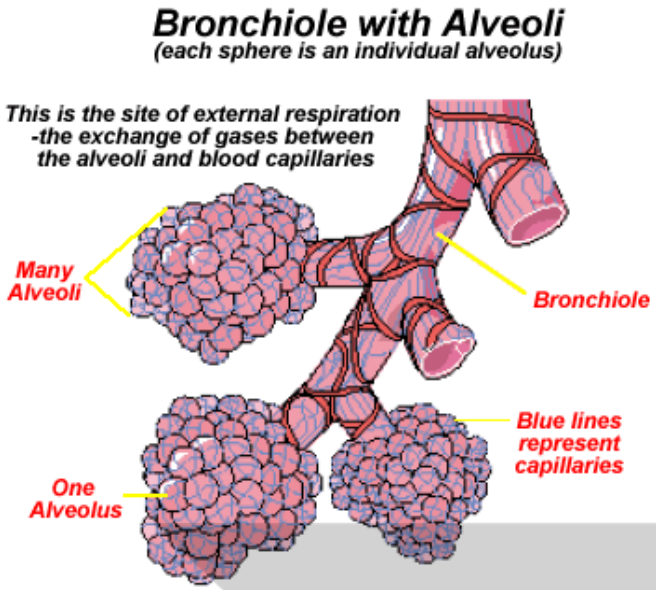
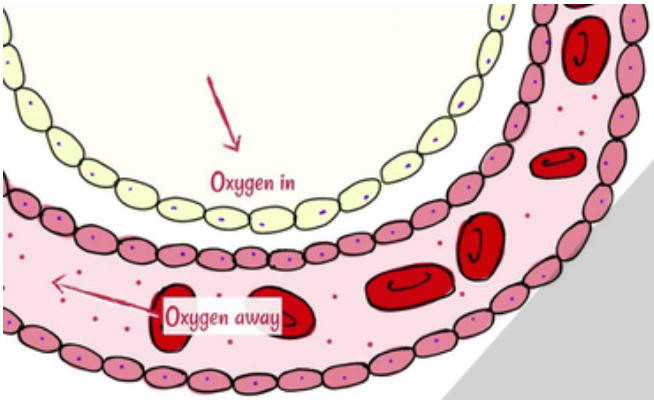


The alveoli are the site for gas exchange between air and blood:



Airway	Number	Approximate diameter	Cartilage	Goblet cells	Smooth muscle	Cilia	Site of gas exchange
trachea	1	1.8 cm	yes	yes	yes	yes	no
bronchus	2	1.2 cm	yes	yes	yes	yes	no
terminal bronchiole	48 000	1.0 mm	no	no	yes	yes	no
respiratory bronchiole	300 000	0.5 mm	no	no	no	a few	no
alveolar duct	9×10^6	400 µm	no	no	no	no	yes
alveoli	3×10^9	250 µm	no	no	no	no	yes

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